Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 10

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Activity 01

```
bisheka@bisheka-VirtualBox:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
bisheka@bisheka-VirtualBox:~$
```

```
bisheka@bisheka-VirtualBox:~$ echo $HOME
/home/bisheka
bisheka@bisheka-VirtualBox:~$
```

```
bisheka@bisheka-VirtualBox:~$ echo $USER
bisheka
bisheka@bisheka-VirtualBox:~$
```

```
bisheka@bisheka-VirtualBox:~$ echo $PWD
/home/bisheka
bisheka@bisheka-VirtualBox:~$
```

```
bisheka@bisheka-VirtualBox:~$ echo $PS1
\[\e]0;\u@\h: \w\a\]${debian_chroot:+($debian_chroot)}\[\033[01;32m\]\u@\h\[\033[00m\]:\[\033[01;34m\]\w\[\033[00m\]\$
bisheka@bisheka-VirtualBox:~$
```

a) No

b) The Unix operating system (shell/environment)

Activity 02

```
bisheka@bisheka-VirtualBox:~$ labclass="This is OSSA Lab Class"
bisheka@bisheka-VirtualBox:~$ echo $labclass
This is OSSA Lab Class
bisheka@bisheka-VirtualBox:~$
```

```
bisheka@bisheka-VirtualBox:~$ echo $num
18
```

```
bisheka@bisheka-VirtualBox:~$ num=`expr 13 \* 5`
bisheka@bisheka-VirtualBox:~$ echo $num
65
```

```
bisheka@bisheka-VirtualBox:~$ num1=`expr $num / 7`
bisheka@bisheka-VirtualBox:~$ echo $num1
9
```

```
bisheka@bisheka-VirtualBox:~$ num1=`expr $num % 7`
bisheka@bisheka-VirtualBox:~$ echo $num1
2
```

```
bisheka@bisheka-VirtualBox:~$ num1=`expr $num1 + 8`
bisheka@bisheka-VirtualBox:~$ echo $num1
10
```

```
#!/bin/bash

name= "Bisheka"
address= "Narahenpita"
country= "Sri Lanka"

echo "Name: $name"
echo "Address: $address"
echo "Country: $country"
```

```
bisheka@bisheka-ubuntu:~$ nano usrinfo
bisheka@bisheka-ubuntu:~$ chmod 777 usrinfo
bisheka@bisheka-ubuntu:~$ ls -l usrinfo
-rwxrwxrwx 1 bisheka bisheka 141 Apr 15 5:20 usrinfo
bisheka@bisheka-ubuntu:~$ bash /usrinfo
Name: Bisheka
Address: Narahenpita
Country: Sri Lanka
bisheka@bisheka-ubuntu:~$
```

```
bisheka@bisheka-ubuntu:~$ chmod 770 usrinfo
bisheka@bisheka-ubuntu:~$ ls -l usrinfo
-rwxrwx--- 1 bisheka bisheka 141 Apr 15 5:20 usrinfo
bisheka@bisheka-ubuntu:~$ bash ./usrinfo
Name: Bisheka
Address: Narahenpita
Country: Sri Lanka
bisheka@bisheka-ubuntu:~$
```

a) bash usrinfo

b) bash ./usrinfo

Activity 03

```
bisheka@bisheka-VirtualBox:~$ mkdir scripts
bisheka@bisheka-VirtualBox:~$ cd scripts
bisheka@bisheka-VirtualBox:~/scripts$ nano script1
```

```
#!/bin/bash

echo "Welcome to SLIIT Server"

echo "Today is : $(date)"
```

```
bisheka@bisheka-VirtualBox:~/scripts$ sh script1
Welcome to SLIIT Server
Today is : Wed Apr 15 05:46:07 AM +0530 2026
```

```
bisheka@bisheka-VirtualBox:~/scripts$ ls -l script1
-rw-rw-r-- 1 bisheka bisheka 72 Apr 15 05:46 script1
```

We can’t type script1 to run because the file does not have execute (x) permission.

```
bisheka@bisheka-VirtualBox:~/scripts$ chmod u+x script1
bisheka@bisheka-VirtualBox:~/scripts$ ./script1
Welcome to SLIIT Server
Today is : Wed Apr 15 05:48:45 AM +0530 2026
bisheka@bisheka-VirtualBox:~/scripts$ ls -l script1
-rwxrw-r-- 1 bisheka bisheka 72 Apr 15 05:46 script1
bisheka@bisheka-VirtualBox:~/scripts$
```